

Do socially (ir)responsible investments pay? New evidence from international ESG data

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ABSTRACT

Using a new dataset of environmental, social and corporate governance (ESG) company ratings and state-of-the-art statistical methodology, this article analyses the performance of socially (ir)responsible investments in the Asia-Pacific region, the United States and Europe. By implementing a variety of portfolio screens on the industry level, our analysis provides the following insights. First, regardless of geographic region, industry or ESG criterion, active selection of high- or low-rated stocks does not provide superior risk-adjusted performance in comparison to passive stock market investments. Second, in the Asia-Pacific region and in the United States, investors concentrating on ethical utility derived from their portfolio choice can follow an ESG-based investment style and still obtain a performance similar to the broad market. However, depending on the industry focus and the ESG criterion that is used, investors in Europe tend to pay a price for socially responsible investing. Third, our results are robust along several dimensions, such as the employed portfolio cut-off rate, the time frame or the consideration of transaction costs.

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1. Introduction

Over the past decade, socially responsible investment (SRI) has become a major trend in the mutual fund industry and a key topic in financial research all around the world.¹ SRI can be broadly defined as an investment process that involves identifying companies with high corporate social responsibility (CSR) profiles where the latter are evaluated on the basis of environmental, social and corporate governance (ESG) criteria (see Renneboog, Ter Horst, & Zhang, 2008a). It implies that investors do not primarily wish to derive financial utility from their investment decisions but also strive for non-financial utility resulting from holding portfolios that are consistent with personal and societal values (see Bollen, 2007).

While the issue of non-financial utility is undisputed, there is a still ongoing debate on the potential economic viability of SRI. In this respect, the literature reveals three opposing views (see Preston & O'Bannon, 1997; Sauer, 1997). The 'doing good while

doing well' view indicates a positive relationship between social and financial performance and suggests superior returns from choosing high-rated stocks. It can be observed if the so-called 'available fund hypothesis' or the 'good management hypothesis' hold. While the former argues that high corporate financial performance yields slack resources enabling firms to invest in socially responsible activities (see Eichholtz, Kok, & Yonder, 2012), the latter implies that meeting the requirements of major stakeholders by ensuring, for example, product enhancement or job security, can lead to higher financial performance as a result of continued business or firm loyalty (see Cornell & Shapiro, 1987; McGuire, Sundgren, & Schneeweis, 1988). The 'doing good but not well' point of view suggests a negative relationship that is linked to the 'managerial opportunism hypothesis' or the 'trade-off theory'. According to the first hypothesis, managers may tend to maximise private gains in prosperous times and placate weak financial performance by increasing the shareholders' welfare through social activities (see Posner & Schmidt, 1992). The second hypothesis asserts that socially responsible activities may siphon off resources from a firm, putting it in relative disadvantage to firms that are less socially active (see Aupperle, Carroll, & Hatfield, 1985). Thus, this line of argument suggests the superiority of low-rated firms. Finally, a last perspective is that SRI neither adds nor destroys portfolio value because CSR is not priced. It resembles the standard framework of finance, where factors that are not proxies for risk do not affect expected returns and socially responsible investors do not reduce

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¹ According to Kempf and Osthoff (2007), in the US alone, almost one out of ten dollars under professional management is invested according to socially responsible principles.

the relative cost of capital to socially responsible companies by favouring their stocks (see [Hamilton, Jo, & Statman, 1993](#)).

Research on the financial benefits of SRI mainly concentrates on comparing the performance of socially responsible mutual funds to the performance of unrestricted benchmark portfolios or conventional funds (see [Renneboog et al., 2008a](#), for a review of this vast literature). The overall findings of these studies covering most markets of the world suggest that the risk-adjusted returns for investing in SRI funds do not appear to be significantly different from conventional fund returns.² That is, there seems to be no performance advantage (or disadvantage) in selecting SRI funds. However, these results should be handled with care because the evaluation of SRI performance on a fund level has several major drawbacks. First, recent evidence indicates that funds classified as SRI funds considerably change their social responsibility status over time because fund managers tend to modify the holdings not only with increases or decreases of social responsibility on the company level but also in response to changes in other criteria or turbulences in the market (see [Wimmer, 2013](#)). SRI funds gradually converge to conventional funds. Thus, an SRI label does not ensure that a fund actually follows pure SRI principles. Second, a number of confounding effects make it difficult to rely upon differences in mutual fund performance to establish the impact of social responsibility screens (see [Kempf & Osthoff, 2007](#); [Sauer, 1997](#)). For example, fund performance does not merely reflect the returns of the underlying securities, but rather, also reflects differences in management fees which can vary widely across fund families and investment objectives. Third, most studies rely on funds' alphas to measure their risk-adjusted performance. This is problematic for several reasons. The first reason is that Jensen's alpha is an appropriate measure of portfolio performance for investors that are well diversified and, therefore, are primarily concerned with their exposure to market (systematic) risk (see [Eling & Schuhmacher, 2007](#)). However, if SRI screens restrict the investment universe, investors may inadvertently subject themselves to otherwise diversifiable (unsystematic) risk. In this case, it is more reasonable to use a performance measure based on total risk, rather than market risk (see [Sauer, 1997](#)). In addition, the use of alternative alphas derived from traditional multifactor models such as those suggested by [Fama and French \(1993\)](#) and [Carhart \(1997\)](#) is still under debate in the finance literature. This is because of a lack of consensus on the appropriate kind and number of model factors (see [Capocci & Hübner, 2004](#)) and certain data snooping biases (see [Lo & MacKinlay, 1990](#)). Finally, statistical inference for the obtained alpha estimates is usually conducted using standard procedures that are based on asymptotic normality, even though, the normality-condition is not fulfilled in typical empirical sample sizes (see [Henry, 2002](#); [Poon & Taylor, 1992](#)). Thus, comparing alphas of SRI and non-SRI portfolios may not be a fully suitable methodology to analyse the impact of SRI on investment performance or at least has to be handled with care.

A smaller strand of the literature deals with specifically constructing SRI portfolios in order to evaluate the advantages of social responsibility screens (see [Derwall, Koedijk, & Ter Horst, 2011](#), for a brief review of the literature). However, the most influential of those studies also have some limitations. First, they primarily use alternative alphas to evaluate investment performance. Second, they concentrate mostly on one dimension of social responsibility (e.g. the eco-efficiency or the directors' board decomposition) or one geographic region (e.g. the United States or Europe) and use different ways to quantify CSR. Thus, their results are not strictly comparable. Third, the subject of corporate governance is usually not covered in studies that analyse more than one dimension

of social responsibility. Fourth, especially in the European case, most studies suffer from very limited sample sizes. Finally, the studies tend to yield contrary results. While, for example, [Derwall, Guenster, Bauer, and Koedijk \(2005\)](#), [Kempf and Osthoff \(2007\)](#) and [Edmans \(2011\)](#) confirm superior performance of certain socially responsible screens, [Brammer, Brooks, and Pavelin \(2006\)](#) and [Hong and Kacperczyk \(2009\)](#) find evidence of negative performance. In other words, the latter studies show higher performance for portfolios of socially least desirable stocks.

The goal of this article is to gain further insights into the impact of social responsibility screens on investment performance. Without being subject to the drawbacks of previous studies, this article makes the following main contributions. First, we analyse SRI on a global scale by not focussing only on the United States and Europe but also taking into account the Asia-Pacific region that has been underanalysed so far. Comparing different parts of the world is of interest because even if the CSR ratings of two companies are identical, the corresponding socially responsible activities may lead to diverse outcomes if the environment (i.e., culture, legal restrictions) are different. Second, we are the first international study to quantify CSR by ESG scores provided by the rating agency Sustainalytics. As they are constructed by rigorous standards, they allow a comparison between geographic regions. Third, we not only analyse ESG scores and geographic regions on an aggregate level. We also consider the three dimensions of corporate social responsibility (E, S and G) separately. Furthermore, we are the first to subdivide our stock sample into industries in order to see whether there are different patterns of SRI performance depending on the industry focus of an SRI strategy. Fourth, we leave the boundaries of an alpha-based performance evaluation by applying the Sharpe ratio.³ This simple and most frequently used reward-to-risk ratio measures the average premium per unit of total risk and therefore is adequate for less than well diversified investors.⁴ It has a decision-theoretic foundation under a wide range of (skewed and fat-tailed) non-normal distributions (see [Schuhmacher & Eling, 2011, 2012](#)) and the significance of performance differences can be tested using state-of-the-art bootstrap methods that are robust to typical features of financial time series like non-normality and serial correlation (see [Ledoit & Wolf, 2008](#)). Fifth, we analyse the robustness of our results along several dimensions. We study the effects of different cut-off rates in our portfolio screens. We then analyse portfolio performance over time. We perform some final checks regarding the application of alternative alphas, the specification of our testing procedure, our portfolio construction, the preparation of the dataset, the choice of risk-free rate and the consideration of transaction costs.

There are basically two types of socially responsible investors (see [Derwall et al., 2011](#)): 'Value-driven investors' (VDI) are concerned merely with the non-financial utility they derive from the SRI attribute (a high ESG rating) of their investment and are willing to accept a loss in financial performance in exchange for that. They can invest in any region and industry covered by our study because there is a sufficient number of high-rated companies to choose from. 'Responsible profit-seekers' (RPS) wish to concentrate on high-rated companies but also gain financial profits from their portfolio choice. They would not invest in regions or industries, where SRI does not provide financial benefits. Besides those

³ Several recent studies evaluate the performance of investment strategies without resorting to alphas (see, for example, [Hatgiannides & Mesomeris, 2007](#); [Szakmary, Shen, & Sharma, 2010](#); [Yao, 2012](#)). However, note that, for completeness, we also apply a standard alpha approach in a robustness check in [Section 4.2](#).

⁴ Note that the Sharpe ratio (and alpha) has the theoretical disadvantage that it can be manipulated technically (see [Goetzmann, Ingersoll, Spiegel, & Welch, 2007](#); [Schuster & Auer, 2012](#)). However, this is not problematic in our application because our portfolio construction is not subject to such manipulation.

² For the exceptions of France, Ireland, Sweden, and Japan, [Renneboog, Ter Horst, and Zhang \(2008b\)](#) show that SRI can lead to significant underperformance.

two responsible types of investors, we can classify another group of investors using ESG scores in their decision process. These are called ‘irresponsible profit-seekers’ (IPS) because they are simply interested in identifying profitable investment opportunities and would even invest in controversial stocks if they showed superior performance. Our study seeks to answer the questions of where (geographic region and industry) and based on what criterion of social responsibility RPS and IPS can invest in order to achieve their goals. To this end, we apply positive screens within each region and industry in order to construct portfolios containing the stocks with the highest ESG ratings and portfolios of stocks with the lowest ratings. We compare the performance of those portfolios to the one of benchmark strategies that simply hold passive portfolios containing all stocks of the relevant subset. This way we can determine whether active selection based on ESG scores can beat the market or eliminate risk-adjusted performance in comparison to the one of passive non-ESG investors. Here, results of the latter kind allow the identification of investment segments in which a VDI has to pay a price for being socially responsible. Additionally, we compare high-rated and low-rated portfolios to each other because we are also interested in providing international evidence regarding the question of whether ‘the good’ significantly outperform ‘the bad’ or vice versa.

Our article is organized as follows. [Section 2](#) describes the main characteristics of our dataset. In particular, it compares ESG ratings worldwide and points out relations to well-known financial variables typically used in portfolio construction. As we are the first to use the ESG dataset of Sustainalytics, our data description is more detailed. Thus, readers that are interested mainly in our core research question can skip this part of the article. [Section 3](#) briefly outlines our portfolio construction procedure and the statistical methodology used for performance comparison. [Section 4](#) presents our main empirical results and the outcomes of our robustness checks. Finally, [Section 5](#) concludes.

2. Data

Our study is based on a new dataset of international ESG scores. The use of ESG scores to quantify corporate social responsibility is particularly interesting because according to [Hill, Ainscough, Shank, and Manullang \(2007\)](#) and [Wimmer \(2013\)](#), relatively few investors have a strict understanding of what exactly they consider to be socially responsible (e.g. not investing in arms industries). Most of them have only a vague idea and base their investment decisions on ESG scores that have become the industry standard for evaluating CSR. That is, ethical portfolios are constructed simply by selecting companies with the highest ESG ratings.

Several rating agencies are specialised in assessing ESG scores.⁵ These agencies evaluate data from various sources (e.g. company filings, media, governments, third-party data providers) on numerous subtopics regarding CSR. Thereby, a large number of subscores are generated, which are eventually condensed into a single composite ESG score for a company. In our study, we obtain ESG data from Sustainalytics, a global leader in sustainability analysis. In contrast to most other data providers, Sustainalytics does not restrict itself to a specific geographic region or the reporting of a composite ESG score. Furthermore, Sustainalytics updates its company evaluations more than once a year. Company profiles are updated for controversial incidents on a monthly basis and ESG scores in the database are adjusted if necessary. Thus, it supplies time

series of separate environmental, social and corporate governance scores (ranging from zero to one hundred) that are constructed on the basis of up to 70 specific indicators (see www.sustainalytics.com) for a wide range of international companies. Environmental scores reflect environmental awareness usually shown by proactive involvement in recycling, waste reduction, environmental cleanup, renewable energies and biotechnology. Social scores comprise issues like labour relations (strong union relationships, employee empowerment, employee profit sharing, good workplace conditions), business practice (R&D, quality assurance, product safety), employment diversity (employment of minorities, women, and disabled persons), and community involvement (sponsoring, charitable donations, education programmes). Finally, governance scores deal with outstanding best practices related to board independence and elections, auditor independence, executive compensation, voting and shareholder rights.

Because of the availability of the ESG scores, our monthly sample ranges from August 2004 to December 2012. It covers large, mid and small capitalisation companies from the Asia-Pacific region (Australia, Bermuda, British Virgin Islands, Cayman Islands, Hong Kong, Japan, New Zealand, Singapore), the United States (including Canada for reasons of grouping) and Europe (Austria, Belgium, Denmark, Finland, France, Germany, Greece, Iceland, Ireland, Italy, Luxembourg, the Netherlands, Norway, Portugal, Spain, Sweden, Switzerland and the United Kingdom). To avoid a possible survivorship bias, we do not exclude companies that were delisted during the sample period. In order to calculate monthly stock returns, we obtain stock prices and dividends (and for further analysis also market capitalisation and book-to-market ratios) for each company from Thomson Reuters Datastream. To handle the well-known inconveniences related to Datastream data, we employ the screening procedures proposed by [Artmann, Finter, Kempf, Koch, and Theissen \(2012\)](#). In each geographic region (Asia-Pacific, United States and Europe), we consider investors that wish to implement ESG-based portfolio screens within the investment universe that is given by all rated stocks in the respective region.⁶ The base currencies in these regions are the Japanese yen, the US dollar and the Euro.⁷ Following [Bauer, Koedijk, and Otten \(2005\)](#), excess returns (over a risk-free rate) required for the computation of the Sharpe ratio are calculated by subtracting 1-month interbank offered rates from the monthly stock returns. Fitting the currencies of the respective regions, we use the interest rates the British Bankers' Association documents for Japan, the United States and Europe.

For the construction of our industry portfolios, we use the industry classification of Sustainalytics. However, as not all specific industries are represented by a sufficient number of companies, we follow [Hyde and Sherif \(2005\)](#) and build four industry sectors based on the companies' output. Specifically, we distinguish between companies producing/offering capital (e.g. automobiles, construction, machinery), consumption (e.g. food producers, retailers, media), financial (e.g. banks, insurance, real estate) and

⁵ Besides Sustainalytics, there are, for example, KLD Research & Analytics (see [Hong & Kacperczyk, 2009; Kempf & Osthoff, 2007](#)), Thomson Reuters (see [Wimmer, 2013](#)), Vigeo (see [Van de Velde, Vermeir, & Corten, 2005](#)) and EIRIS (see [Brammer et al., 2006](#)).

⁶ We cannot perform our analysis at the country level because not all countries are represented with sufficient numbers of companies that would allow a reasonable portfolio construction.

⁷ As in [Grobys \(2014\)](#), for some stocks a currency conversion is necessary that we base on WM/Reuters exchange rates. Although the performance of international portfolios is generally subject to exchange rate fluctuations, they do not strongly influence our analysis because only a relatively small number of stocks is subject to the conversion. Note that we explicitly do not choose the US dollar as the base currency for all regions because then our portfolio performance would not simply reflect the performance of the underlying stocks but also the performance on the currency market (see [Glen & Jorion, 1993](#)).

other/miscellaneous (e.g. chemicals, pharmaceuticals, commodities) goods/services.⁸

Table 1 reports basic descriptive statistics for our sample subdivided by geographic regions and industry sectors. Besides the number of stocks within each subset, Panel A shows the cross-sectional means of the time-series averages (and of the time-series standard deviations in parentheses) for the stocks' excess returns, market capitalisations and book-to-market ratios. Panel B displays similar means for the environmental, social and corporate governance scores as well as for the total company score that is calculated as the equally weighted average of the three individual scores.

Our sample covers 632, 914 and 572 companies from the Asia-Pacific region, the United States and Europe, respectively. The largest numbers of firms can be observed in the sectors for consumption and miscellaneous goods/services. Turning to the financial data, we find that even though our sample period covers the US subprime crisis and the European debt crisis, mean excess returns in all regions and industries are positive (see also Fama & French, 2012). Mean excess returns (volatilities) tend to be lower (higher) in the Asia-Pacific region than in the United States and Europe resembling the fact that Asia-Pacific markets are known to experience more frequent and longer lasting financial downturns than the United States or Europe and additionally suffer from international crisis-spillovers (see Hong, Lee, & Tang, 2010). In the Asia-Pacific region, mean excess returns (market values per company) take their highest value in the consumption (financial) sector, whereas the miscellaneous sector is the dominant one in the United States and Europe. As far as the book-to-market ratios are concerned, we can observe the common result that they tend to be the highest in the financial sectors of all regions (see Lewellen, 1999).

A closer look at the descriptive statistics of our ESG scores provides some interesting insights. First, in all regions and industries, companies are rated higher in terms of governance issues than they are in environmental and social issues. With one exception, social ratings also tend to be higher than environmental ratings. Second, while Europe is the region with the on average highest rated companies in the environmental and social areas, the United States show leading ratings in terms of corporate governance. Third, in the Asia-Pacific region and in Europe, we can observe the highest standard deviations in the environmental area indicating more drastic changes in ratings than in social and governance issues. Fourth, standard deviations are highest in Europe implying a more turbulent process of change than in the Asia-Pacific region and the United States. Finally, in an industry comparison for the Asia-Pacific region, the highest environmental and social ratings can be observed in the capital sector, while the highest governance scores are found in the consumption sector. In the United States, environmental and governance ratings are highest in capital sector, while the financial sector exhibits the highest social ratings. In Europe, the consumption sector has the highest environmental and social ratings, while companies with superior governance tend to be found in the miscellaneous sector.

Before turning to the methodology and the empirical results of our study, it is instructive to analyse the relationships between ESG-based stock selection and other selection procedures whose profitability has been highlighted in previous research. This is important because newly proposed investment strategies may actually select the same stocks as classic strategies and thus the costs of information collection for the new strategy may be avoided (see Bali, Cakici, & Whitelaw, 2011; Novy-Marx, 2012). Four major strategies have been identified in the literature and build key explanatory variables in modern multi-factor models.

⁸ A detailed list of the industries assigned to each sector is available from the authors upon request.

The best-known are arguably the ones based on company sizes and book-to-market ratios. They imply that stocks with low market capitalisation (high book-to-market ratios) outperform stocks with high market capitalisation (low book-to-market ratios) (see Liew & Vassalou, 2000). More recent research concentrates on momentum- and volatility-based strategies. Momentum approaches are trend-following strategies that select stocks based on the idea that winners (losers) of the recent past, i.e., stocks with the highest (lowest) returns in a given period of the past, continue their growth (decline) in the near future (see Grobys, 2014; Yao, 2012). Volatility-based strategies make use of the empirical observation that there seems to be the tendency that low-volatility stocks have higher returns than highly volatile ones (see Baker, Bradley, & Wurgler, 2011; Dutt & Humphery-Jenner, 2013).

Similar to Bali et al. (2011), Table 2 reports the time averages of cross-sectional correlations between ESG scores and between the ESG scores and the well-known portfolio selection strategies, i.e., the variables needed for their implementation. As noted above, the market capitalisation and book-to-market values stem from Thomson Reuters Datastream. Momentum selection is represented by a simple strategy using the compounded excess returns over the prior 12 months, skipping the most recent month.⁹ Volatilities are estimated by means of rolling standard deviations, where the incorporated data covers the previous 12 months.¹⁰

In Panel A, analysing the relationship between the ESG scores, we can detect high correlations of up to about 0.80 between the environmental and social scores. This indicates that portfolio selection based on these scores may yield similar risk-adjusted performance. In contrast, correlations between those two scores and the governance score show a different picture. While they are high in the financial sectors of the Asia-Pacific region and Europe, they are lower in other industries and particularly low in the United States. Finally, as to the nature of their construction, the company scores are highly correlated with the individual ESG scores.

Panel B indicates that (when Sustainalytics data is used) ESG-based portfolio selection does not resemble classic investment strategies and thus can be considered as an independent portfolio selection criterion.¹¹ Correlations between the ESG scores and the alternative criteria are low and economically insignificant. They remain below absolute values of 0.39 and 0.29 in the case of market capitalisation and book-to-market ratios, respectively. In the case of momentum, they hardly deviate from zero. A relation to volatility-investing is also very unlikely because correlations never breach the absolute level of 0.29.

3. Methodology

3.1. Portfolio construction

In order to evaluate whether high- or low-rated ESG portfolios outperform typical passive benchmarks or each other, we construct and compare a series of portfolios based on Sustainalytics ESG scores. A high-rated portfolio is obtained as follows. At the end of each month, we rank the stocks in the relevant stock universe (all stocks or a specific industry in a given geographic region) according to their environmental, social, governance or company scores. Based on this ranking, we then select the best 5% of the stocks

⁹ Following Yao (2012), we also analysed the relation to short-run (intermediate-term) momentum based on the months $t-6$ to $t-2$ ($t-12$ to $t-7$) and found similar results.

¹⁰ However, we could also do otherwise because Dutt and Humphery-Jenner (2013) point out that the size of the window used in the rolling estimation of standard deviations does not influence the presence of the low-volatility anomaly.

¹¹ This result is a close analogon to the very weak dependence Keef and Khaled (2012) find among selected seasonal stock market effects.

Table 1
Descriptive statistics.

	Asia-Pacific					United States					Europe				
	Total	Cap	Con	Fin	Misc	Total	Cap	Con	Fin	Misc	Total	Cap	Con	Fin	Misc
<i>Panel A: Stock data</i>															
Number	632	137	179	126	190	914	186	307	175	246	572	131	177	123	141
Excess return	0.48 (10.69)	0.34 (11.02)	0.68 (9.52)	0.66 (10.67)	0.28 (11.55)	0.72 (9.35)	0.74 (10.41)	0.73 (8.61)	0.37 (9.35)	0.94 (9.49)	0.57 (8.76)	0.59 (10.04)	0.63 (7.80)	0.32 (8.94)	0.68 (8.63)
Market value	880.62 (243.33)	842.12 (245.36)	964.30 (223.10)	1060.39 (337.91)	710.33 (198.19)	16.05 (3.98)	11.91 (3.64)	13.77 (3.15)	13.22 (3.23)	24.05 (5.81)	11.92 (3.20)	7.92 (2.67)	12.84 (3.38)	12.56 (3.71)	13.91 (3.01)
Book-to-market	0.78 (0.20)	0.75 (0.21)	0.70 (0.17)	1.00 (0.27)	0.73 (0.19)	0.53 (0.17)	0.55 (0.28)	0.44 (0.12)	0.81 (0.23)	0.41 (0.11)	0.65 (0.21)	0.61 (0.24)	0.48 (0.15)	1.02 (0.35)	0.56 (0.16)
<i>Panel B: ESG data</i>															
Environmental	48.64 (6.47)	52.80 (8.66)	48.86 (5.18)	42.06 (6.21)	49.78 (6.28)	46.06 (5.61)	49.15 (5.90)	47.25 (5.57)	40.76 (5.35)	46.00 (5.62)	54.94 (7.04)	56.69 (7.57)	57.19 (6.97)	49.69 (7.48)	55.07 (6.24)
Social	50.32 (4.61)	51.03 (4.81)	49.97 (4.63)	49.40 (4.95)	50.74 (4.23)	50.33 (4.42)	50.96 (4.21)	50.27 (4.62)	51.20 (4.25)	49.31 (4.47)	56.39 (6.04)	56.80 (6.44)	57.55 (6.15)	54.19 (6.21)	56.48 (5.40)
Governance	58.41 (4.87)	59.48 (5.39)	60.19 (4.50)	52.34 (4.71)	59.98 (4.94)	65.89 (5.71)	69.39 (6.03)	68.33 (5.12)	56.52 (5.77)	66.88 (6.15)	63.97 (6.21)	64.67 (6.73)	65.73 (6.39)	57.81 (5.92)	66.49 (5.77)
Company	52.45 (3.65)	54.44 (4.16)	53.01 (3.58)	47.94 (3.91)	53.50 (3.18)	54.09 (3.40)	56.50 (3.33)	55.28 (3.73)	49.50 (3.01)	54.06 (3.30)	58.43 (5.07)	59.39 (5.43)	60.16 (5.18)	53.90 (5.25)	59.35 (4.43)

For each geographic region and subdivided by industries producing/offering capital (cap), consumption (con), financial (fin) and miscellaneous (misc) goods/services, this table reports key properties of our stock market sample in the period from August 2004 to December 2012. Panel A displays the numbers of stocks within each subgroup. Furthermore, it shows the cross-sectional means of the time-series averages (and of the time-series standard deviations in parentheses) for monthly percentage excess-returns, market capitalisations (in billions of Japanese yen, US dollars and Euro, respectively) and book-to-market ratios. Financial data is obtained from Thomson Reuters Datastream. Interbank offered rates documented by the British Bankers' Association are used for the calculation of excess returns. Panel B characterises the CSR of our sample companies. Similar to Panel A, it reports the means of the Sustainalytics environmental, social, corporate governance and total company scores (with their standard deviations in parentheses). The company score is obtained as an equal weighted average of the three individual ESG scores.

and form an equally weighted portfolio. This portfolio is held until changes in the ratings make it necessary to restructure its decomposition. This procedure yields a total of 60 portfolios (3 regions $\times$ 5 sectors $\times$ 4 ESG criteria) with time series of monthly excess returns from September 2004 to December 2012. As the cut-off rate may influence our results, we repeat this portfolio construction by considering portfolios concentrating on the top 10%, 15%, 20% and 25% of the stocks. This increases of number of high-rated ESG portfolios

to 300 (60 portfolios $\times$ 5 cut-off rates). Additional 300 portfolios of socially less desirable stocks are obtained by focussing on the securities with the lowest ratings. The benchmark portfolios are simple equally weighted portfolios comprising all stocks of the relevant stock universe. Thus, they represent typical passive investments in broad stock market or industry indices. This choice of benchmark is motivated by the fact that in practice SRI funds typically compare themselves with those kinds of benchmarks in their

Table 2
Correlations between ESG scores and classic portfolio selection variables.

	Asia-Pacific					United States					Europe				
	Total	Cap	Con	Fin	Misc	Total	Cap	Con	Fin	Misc	Total	Cap	Con	Fin	Misc
<i>Panel A: ESG variables</i>															
E-S	0.66	0.68	0.69	0.78	0.64	0.62	0.77	0.59	0.50	0.64	0.71	0.69	0.70	0.74	0.68
E-G	0.41	0.40	0.31	0.61	0.31	0.26	0.28	0.18	0.27	0.20	0.48	0.41	0.47	0.60	0.38
S-G	0.53	0.53	0.54	0.64	0.46	0.33	0.38	0.38	0.23	0.29	0.57	0.51	0.60	0.61	0.56
E-C	0.87	0.88	0.85	0.90	0.84	0.84	0.89	0.80	0.81	0.84	0.86	0.83	0.84	0.90	0.82
S-C	0.86	0.86	0.90	0.89	0.85	0.83	0.90	0.85	0.73	0.84	0.88	0.87	0.89	0.87	0.89
G-C	0.75	0.72	0.70	0.86	0.71	0.66	0.62	0.64	0.68	0.61	0.81	0.79	0.81	0.85	0.79
<i>Panel B: Traditional variables</i>															
MV-E	0.06	0.02	0.10	0.25	-0.06	0.18	0.38	0.16	0.30	0.25	0.21	0.17	0.25	0.31	0.23
MV-S	0.06	-0.03	0.10	0.24	-0.03	0.12	0.36	0.06	0.06	0.17	0.24	0.20	0.23	0.33	0.29
MV-G	0.00	-0.08	0.05	0.11	-0.02	0.06	0.21	0.03	-0.15	0.09	0.21	0.06	0.18	0.31	0.29
MV-C	0.05	-0.02	0.11	0.22	-0.05	0.16	0.39	0.12	0.10	0.23	0.26	0.17	0.26	0.36	0.33
BTM-E	0.01	0.20	0.18	-0.26	0.04	-0.10	-0.07	-0.03	-0.01	-0.08	-0.04	0.13	0.06	-0.01	0.19
BTM-S	0.00	0.10	0.09	-0.23	0.00	-0.08	-0.09	0.00	-0.02	-0.06	-0.03	0.07	0.05	-0.01	0.18
BTM-G	-0.12	0.05	0.09	-0.28	-0.19	-0.13	-0.14	0.08	-0.14	-0.03	-0.11	-0.04	0.07	-0.03	-0.05
BTM-C	-0.03	0.16	0.15	-0.29	-0.06	-0.13	-0.12	0.01	-0.09	-0.06	-0.07	0.06	0.08	-0.02	0.12
MOM-E	-0.01	-0.02	-0.02	0.01	-0.01	0.00	0.00	-0.01	-0.01	-0.02	-0.02	-0.03	-0.02	-0.02	-0.03
MOM-S	-0.01	-0.02	-0.01	0.01	0.00	-0.01	-0.01	-0.01	-0.01	-0.01	-0.03	-0.03	-0.03	-0.05	-0.03
MOM-G	0.01	-0.01	0.00	0.03	0.02	0.01	0.01	-0.01	0.02	0.01	-0.01	-0.03	-0.03	-0.02	-0.02
MOM-C	-0.01	-0.02	-0.01	0.02	0.01	0.00	0.00	-0.01	0.00	-0.01	-0.02	-0.04	-0.03	-0.03	-0.03
VOLA-E	-0.08	0.01	-0.11	-0.05	-0.16	-0.06	0.07	-0.20	0.04	-0.14	-0.07	0.04	-0.16	0.01	-0.13
VOLA-S	-0.03	0.05	-0.16	0.02	-0.06	-0.08	0.05	-0.25	-0.06	-0.09	-0.11	-0.02	-0.19	-0.08	-0.14
VOLA-G	0.01	0.03	-0.05	0.13	0.03	-0.03	-0.01	-0.22	0.15	0.02	-0.04	0.00	-0.17	0.01	0.01
VOLA-C	-0.05	0.02	-0.14	0.04	-0.09	-0.07	0.05	-0.29	0.07	-0.10	-0.09	0.01	-0.21	-0.01	-0.10

Panel A of this table shows the time averages of the cross-sectional correlations between the Sustainalytics ESG scores, i.e., the environmental (E), social (S), corporate governance (G) and company (C) scores. Similarly, Panel B characterises the relationship between the ESG scores and other classic portfolio selection variables used in practice and financial research. Market value (MV) and book-to-market (BTM) ratios are obtained from Thomson Reuters Datastream. Momentum (MOM) is evaluated on the basis of excess returns in the previous 12 months, excluding the most recent month. Volatilities (VOLA) are estimated by means of 12-months rolling-window standard deviations. Industry abbreviations are used as in Table 1.

fund brochures.¹² Furthermore, it allows to simultaneously analyse the issue that SRI screens impose a constraint on the available investment universe and may therefore limit the diversification possibilities and consequently shift the mean–variance frontier towards less favourable risk–return tradeoffs than those of conventional portfolios (see Renneboog et al., 2008a; Sauer, 1997).

This portfolio construction enables us to obtain portfolios that merely reflect the performance of their underlying securities of socially screened stocks. Unlike most SRI mutual funds, our portfolios also ensure a high degree of ESG persistence. This can be seen in Fig. 1 showing the development of the mean ratings for the screened portfolios that are based on the entire samples of stocks in the Asia-Pacific region, the United States and Europe, respectively.¹³ It also provides some additional interesting insights. First, in all regions low-rated companies appear to be undergoing continuous improvement in all areas of CSR. According to Lucey and Zhang (2010), this observation can have its cause in financial integration proceeding to higher levels over time. Second, while the environmental scores of high-rated US and European companies remain largely unchanged or show only slight downward tendencies, we can observe a distinct decline in the ratings of Asia-Pacific companies that may be linked to the reckless expansion of production capacities by some Asian corporations (see, for example, Hung, 2008). Third, the highest-rated US companies appear to lose their superior corporate governance status over time. This is not the effect of new companies entering the sample over time and can be backed up by regressing governance scores on time.

3.2. Performance hypothesis testing with the Sharpe ratio

We measure the risk-adjusted performance of a stock portfolio by its Sharpe ratio. For a portfolio with the T excess returns $r_1, \dots, r_T$, it is calculated as the ratio of the mean excess return $\hat{\mu} = (r_1 + \dots + r_T)/T$ to the standard deviation of excess returns $\hat{\sigma} = \{[(r_1 - \hat{\mu})^2 + \dots + (r_T - \hat{\mu})^2]/(T-1)\}^{0.5}$, i.e., $\hat{\mu}/\hat{\sigma}$. As empirically observed Sharpe ratios are merely estimates of the true portfolio Sharpe ratios, statistical hypothesis testing is required to reasonably compare the Sharpe ratios of two portfolios. In order to conduct reliable statistical inference, we do not apply the traditional test of Jobson and Korkie (1981) and Memmel (2003) because it is not robust to non-normality, autocorrelation and typical small sample biases (see Auer & Schuhmacher, 2013). Instead, we use a bootstrap test recently proposed by Ledoit and Wolf (2008), hereafter LW.¹⁴ This new statistical procedure tests the null hypothesis of equal risk-adjusted performance, i.e., $H_0: \Delta = \eta_i - \eta_j = 0$, where η_i and η_j are the true Sharpe ratios of two portfolios i and j .

Based on the vast literature documenting the improved inference accuracy of the studentised bootstrap over standard inference making use of asymptotic normality,¹⁵ LW propose to construct a symmetric studentised time series bootstrap confidence interval for a given error probability α and to declare two Sharpe ratios different if zero (the hypothetical value of the Sharpe ratio difference) is not contained in the obtained interval or if α lies below the corresponding p -value. To this end, they approximate the two-sided

distribution function of the studentised difference via a distribution obtained from M bootstrap resamples, i.e., $F[\hat{\Delta} - \Delta]/s(\hat{\Delta}) \approx F[\hat{\Delta}^* - \hat{\Delta}/s(\hat{\Delta}^*)]$, where Δ is the true difference between the Sharpe ratios, $\hat{\Delta}$ is the estimated difference computed from the original data, $s(\hat{\Delta})$ is a standard error for $\hat{\Delta}$ (also obtained from the original data), $\hat{\Delta}^*$ is the estimated difference computed from bootstrap data, and $s(\hat{\Delta}^*)$ is a standard error for $\hat{\Delta}^*$ (also obtained from bootstrap data). At a nominal level of α , the confidence interval is then constructed as $\hat{\Delta} \pm z_{1-\alpha/2}^* s(\hat{\Delta})$, where $z_{1-\alpha/2}^*$ denotes the $(1-\alpha)$ -quantile of $F[\hat{\Delta}^* - \hat{\Delta}/s(\hat{\Delta}^*)]$. LW point out that with heavy-tailed data or data of time series nature, this quantile will typically be somewhat larger than its standardnormal counterpart (used in the traditional test) in small to moderate samples, resulting in more conservative inference.

To calculate numerical values of the confidence interval, LW specify the particular bootstrap method and the way of computing $s(\hat{\Delta})$ and $s(\hat{\Delta}^*)$ as follows. To generate bootstrap data, they use the circular block bootstrap of Politis and Romano (1992), involving resampling blocks of pairs from the observed excess return pairs (r_{ti}, r_{tj}) , $t = 1, \dots, T$, with replacement. To conduct this bootstrap, one can either use the LW calibration procedure for selecting the required fixed block size b (see Ledoit & Wolf, 2008, sec. 3.2.2) or follow recent literature that selects it manually to preserve even fine autocorrelation structure in the data (see Behr, Guettler, & Truebenbach, 2012). We follow the latter and set $b = 5$. Given the computational power of modern computers, LW suggest that a number of $M = 5000$ bootstrap sequences is sufficient for reliable inference. We follow this suggestion. Next, turning to the standard errors, $s(\hat{\Delta})$ is calculated based on HAC kernel estimation. In detail, LW propose to use the prewhitened quadratic spectral kernel of Andrews and Monahan (1992). The standard error $s(\hat{\Delta}^*)$ is the natural standard error computed from the bootstrap data, making use of the special block dependence structure (see Götze & Künsch, 1996).

4. Main results

4.1. Performance evaluation

To get a first impression of SRI performance and of the magnitude of differences to the benchmarks, Table 3 presents the results for a portfolio cut-off rate of 5%. Besides the percentage means, standard deviations and Sharpe ratios of our portfolios that concentrate on the 5% firms with the highest/lowest ESG ratings, we also report the same statistics for the benchmark portfolios' excess returns.

In a comparison of the screened portfolios to the benchmarks, we find that, in only 15 of 60 cases, high-rated portfolios show higher Sharpe ratios than their respective benchmarks. In the Asia-Pacific region, we find this kind of outperformance for social screens (in the consumption sector), governance screens (in the total universe and the capital and financial sectors) and composite company rating screens (in the financial and miscellaneous sectors). In the United States, outperformance can be detected for the environmental screens (total, consumption, financial), the social screens (consumption, financial, miscellaneous), the governance screens (financial) and the screens based on the company rating (consumption, financial). Finally, in Europe, this occurs for no ESG criterion or sector.

In contrast, low-rated portfolios outperform their benchmarks in 34 of 60 cases.¹⁶ In the Asia-Pacific region, environmental screens

¹² See, for example, the '2012 Report on Sustainable and Responsible Investing Trends in the United States' and the 'Global Sustainable Investment Review 2012' available under www.ussif.org and www.gsi-alliance.org, respectively.

¹³ Similar results are obtained for the industry portfolios. The corresponding figures are available from the authors upon request. In addition, note that ESG persistence does not mean ESG constancy. Of course, portfolio ratings change over time because companies are reevaluated on a regular basis.

¹⁴ For recent applications, see Scherer (2011), Kroencke and Schindler (2012), Thornton and Valente (2012) and Cumming, Haß, and Schweizer (2013). The source code of the test is available under www.econ.uzh.ch/faculty/wolf/publications.html.

¹⁵ See, for example, Hall (1992) for i.i.d. data and Lahiri (2003) for time series data.

¹⁶ Note that this count is based on the exact Sharpe ratios and not on the rounded numbers of Table 3. Unlike the table suggests, there are no cases of equal Sharpe ratios.

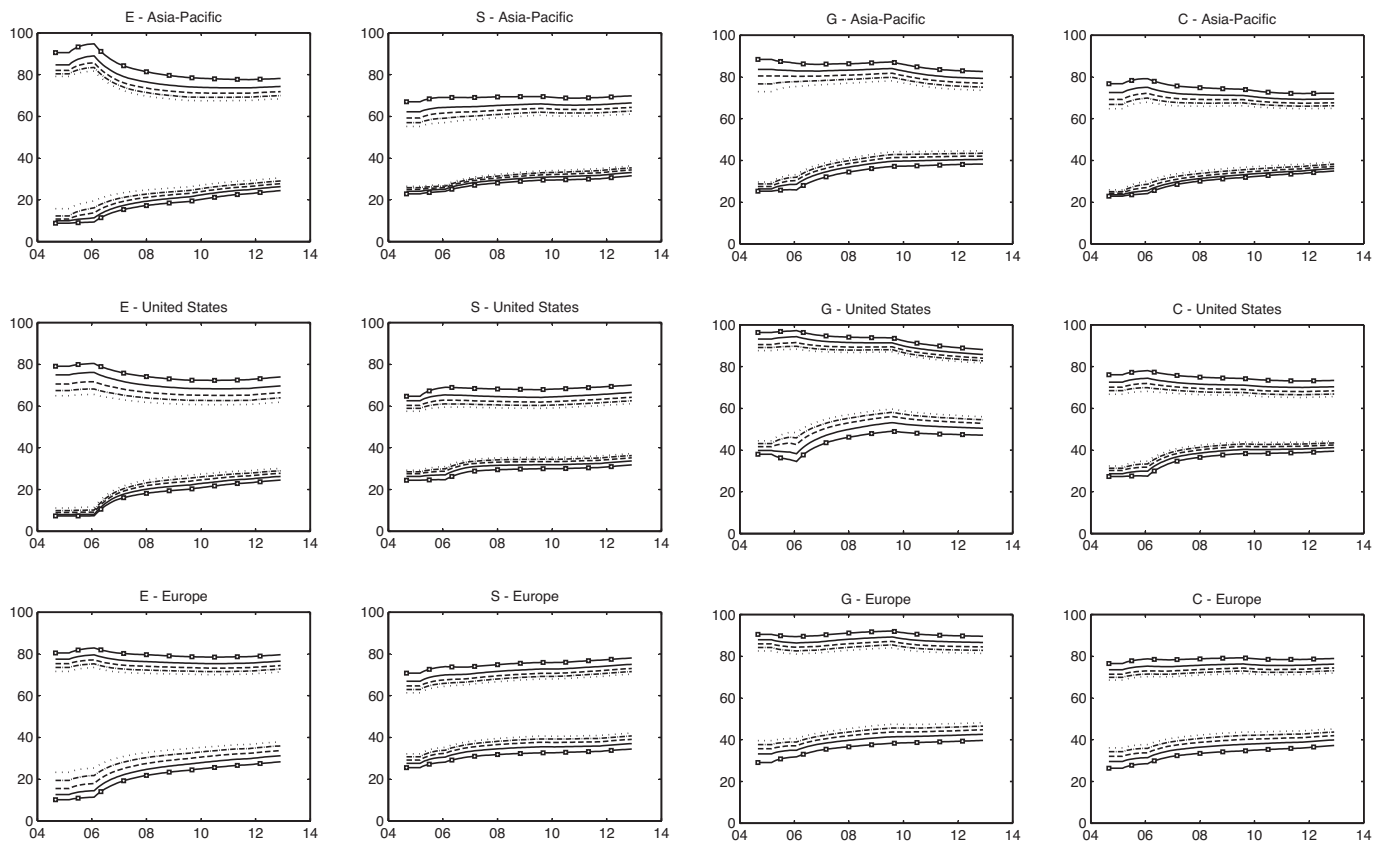


Fig. 1. ESG portfolio ratings over time. For each month in our sample period, the upper (lower) half of each graph in this figure shows the mean ESG scores of portfolios containing the best-rated (worst-rated) firms in the respective region. Portfolios with 5%, 10%, 15%, 20% and 25% cut-off rates are represented by squared-solid, solid, dashed, dashed-dotted and dotted lines, respectively.

(all sectors), social screens (total, capital, miscellaneous), governance screens (total) and company screens (all sectors except for consumption) show this superior performance. In the United States, we can observe it for environmental screens (financial), social screens (all sectors except financial) and company screens (capital). In Europe, environmental and social screens lead to outperformance in all sectors. The outperformance of governance screens (capital, consumption, miscellaneous) and company screens (total, capital) seems to be more sector-limited.

Evaluating the relative performance of the screened portfolios, we find that high-rated ones tend to outperform their low-rated counterparts only in 18 of 60 cases. In the Asia-Pacific region, social screens (consumption, financial), governance screens (total, capital, financial) and screens using the company score (consumption, financial) show evidence of outperformance. In the United States, higher Sharpe ratios can be observed when environmental screens (total, consumption, financial), social screens (financial, miscellaneous), governance screens (total, capital, financial) and screens based on the company score (total, consumption, financial) are applied. In Europe, no outperformance of high- over low-rated portfolios can be detected.

As these initial results do not consider the statistical significance of the detected performance differences and might be driven by the specific cut-off rate, Table 4 reflecting the key results of our empirical analysis chooses a more suitable visualisation of results that also allows to identify potentially existing patterns more clearly. In detail, it compares our 300 high-rated and our 300 low-rated portfolios to their respective benchmarks and reports the signs (+ and –) and significances (a, b and c for levels of 1%, 5% and 10%,

respectively) of the resulting Sharpe ratio differences.¹⁷ Similarly, the low-rated portfolios are compared to the high-rated ones. Thus, for example, the entry c+ for the 25% low-to-high environmental comparison in the capital sector of the Asia-Pacific region implies that the 25% low-rated (high-rated) stocks significantly underperform (outperform) the 25% high-rated (low-rated) stocks at a significance level of 10%.

At first glance, we can see that, in the Asia-Pacific region, none of the previously detected performance differences are statistically significant. Furthermore, except for five specific screens at cut-off rates of 20% and 25% in the capital sector, high-rated and low-rated portfolios neither significantly differ from their benchmarks nor from each other. Two of those cases show low-rated portfolios outperforming their benchmarks in environmental screens, while the remaining three indicate low-rated stocks outperforming their high-rated counterparts when environmental and composite company scores are used in the screening process. Thus, the overall picture implies that investors in the Asia-Pacific region tend to have no significant financial advantage or disadvantage from an (un)ethical portfolio choice. This makes the region rather uninteresting for RPS and IPS but still suitable for VDI. They can build portfolios that are consistent with their personal value systems and beliefs without sacrificing financial performance in comparison to the benchmark.

¹⁷ In our application, it does not matter whether the LW test is performed in a one- or a two-sided form. Both approaches lead to the same conclusions regarding the significance of differences.

Table 3
Performance of ESG-based selection at a cut-off rate of 5%.

	Asia-Pacific					United States					Europe				
	Total	Cap	Con	Fin	Misc	Total	Cap	Con	Fin	Misc	Total	Cap	Con	Fin	Misc
<i>Benchmark</i>															
Mean	0.33	0.24	0.38	0.43	0.33	0.51	0.61	0.49	0.28	0.63	0.52	0.80	0.46	0.18	0.68
Standard deviation	5.81	6.93	4.65	6.89	5.80	5.60	7.07	4.94	7.04	5.26	5.34	6.66	4.05	6.65	5.34
Sharpe ratio	0.06	0.03	0.08	0.06	0.06	0.09	0.09	0.10	0.04	0.12	0.10	0.12	0.11	0.03	0.13
<i>High portfolios</i>															
<i>Environmental</i>															
Mean	0.13	0.19	0.07	0.46	−0.03	0.49	0.15	0.55	0.59	0.33	0.06	0.15	0.27	0.20	0.16
Standard deviation	6.35	7.98	5.96	7.66	6.12	4.90	7.60	3.75	8.91	6.01	5.28	7.87	5.00	8.31	6.62
Sharpe ratio	0.02	0.02	0.01	0.06	0.00	0.10	0.02	0.15	0.07	0.05	0.01	0.02	0.05	0.02	0.02
<i>Social</i>															
Mean	0.19	0.07	0.67	0.31	0.31	0.47	0.27	0.41	0.53	1.05	0.24	0.49	0.01	−0.24	0.30
Standard deviation	6.35	8.76	5.51	7.58	7.36	5.30	7.85	3.65	7.91	6.92	5.27	7.59	4.03	8.86	7.24
Sharpe ratio	0.03	0.01	0.12	0.04	0.04	0.09	0.03	0.11	0.07	0.15	0.05	0.06	0.00	−0.03	0.04
<i>Governance</i>															
Mean	0.57	0.42	0.27	1.05	0.11	0.31	0.51	0.36	0.78	0.20	0.11	−0.03	−0.20	−0.45	0.50
Standard deviation	7.35	9.12	6.14	8.62	8.21	5.86	7.67	4.87	11.61	6.45	5.36	7.38	4.46	9.22	6.74
Sharpe ratio	0.08	0.05	0.04	0.12	0.01	0.05	0.07	0.07	0.07	0.03	0.02	0.00	−0.04	−0.05	0.07
<i>Company</i>															
Mean	0.26	0.07	0.43	0.61	0.78	0.30	0.08	0.49	0.63	0.17	0.14	0.01	0.02	−0.20	0.28
Standard deviation	6.49	8.70	5.55	7.79	7.51	5.14	7.82	3.81	10.23	6.32	5.28	7.65	4.08	9.35	6.72
Sharpe ratio	0.04	0.01	0.08	0.08	0.10	0.06	0.01	0.13	0.06	0.03	0.03	0.00	0.01	−0.02	0.04
<i>Low portfolios</i>															
<i>Environmental</i>															
Mean	0.43	0.53	0.68	0.58	0.61	0.46	0.28	0.45	0.36	0.70	0.65	0.95	0.93	0.34	0.98
Standard deviation	6.61	8.19	6.97	6.53	6.95	5.44	7.02	7.47	6.24	7.33	6.30	7.23	8.06	6.96	6.52
Sharpe ratio	0.06	0.07	0.10	0.09	0.09	0.08	0.04	0.06	0.06	0.10	0.10	0.13	0.12	0.05	0.15
<i>Social</i>															
Mean	0.47	0.59	0.13	0.00	0.47	0.77	0.69	1.01	0.11	0.96	0.95	1.09	1.00	0.33	1.16
Standard deviation	5.97	6.82	6.51	6.37	7.29	5.71	7.78	6.23	6.21	7.10	6.20	7.52	6.38	6.50	6.99
Sharpe ratio	0.08	0.09	0.02	0.00	0.06	0.13	0.09	0.16	0.02	0.14	0.15	0.14	0.16	0.05	0.17
<i>Governance</i>															
Mean	0.34	0.07	0.32	0.01	0.15	0.16	0.33	0.52	−0.19	0.60	0.47	1.11	0.66	0.04	0.87
Standard deviation	5.75	6.77	5.74	6.25	6.26	5.52	8.97	6.14	5.36	5.96	5.59	7.41	5.16	6.15	6.37
Sharpe ratio	0.06	0.01	0.06	0.00	0.02	0.03	0.04	0.09	−0.04	0.10	0.08	0.15	0.13	0.01	0.14
<i>Company</i>															
Mean	0.36	0.48	0.35	0.46	0.89	0.28	0.84	0.50	−0.08	0.71	0.69	1.31	0.71	0.20	0.66
Standard deviation	5.90	7.10	6.28	6.38	6.37	5.12	8.54	6.14	5.09	6.23	5.97	7.82	7.16	8.07	7.20
Sharpe ratio	0.06	0.07	0.06	0.07	0.14	0.05	0.10	0.08	−0.02	0.11	0.12	0.17	0.10	0.02	0.09

For a selection of benchmark and socially screened portfolios, this table shows the percentage time-series means, standard deviations and Sharpe ratios of excess returns. The benchmark portfolios comprise all stocks of the respective geographic regions or industries. High (low) portfolios contain the 5% best (worst) stocks according to a specific ESG criterion. Within all portfolios, stocks are weighted equally. Portfolio decompositions are adjusted at the end of every month provided that changes in the ESG ratings have occurred. Abbreviations are used as in Table 1. The Sharpe ratios of portfolios that are higher than their respective benchmarks are marked in bold print.

In the United States, a slightly different but still weak pattern emerges. Across all cut-off rates, sectors and ESG criteria, only seven significant differences can be observed. Two of them indicate the higher performance of low-rated stocks over high-rated ones when a 25% environmental cut-off in the miscellaneous sector is employed and a higher performance of low-rated stocks in comparison to the benchmark for a 10% social cut-off in the consumption sector. The other five concentrate on the miscellaneous sector and show high-rated portfolios significantly underperforming their benchmarks when environmental screens (cut-off rate 25%), governance screens (5% and 15%) and company screens (5% and 20%) are employed. However, as these results are bound to very specific cut-off rates, they cannot be considered as systematic effects. This leads to a similar conclusion as in the Asia-Pacific region. Again, the region is more suitable for VDI than it is for RPS and IPS. These results are consistent with the overall evidence from the US SRI-fund literature documented by Renneboog et al. (2008a, 2008b).

Switching our attention to Europe, we find evidence that a VDI of this geographic region has to sacrifice financial performance

in order to construct certain socially responsible portfolios. If we define evidence to be strongest for cases where at least four cut-off rates in a sector provide significant results pointing in the same direction, we can identify eight scenarios where high-rated portfolios significantly underperform their benchmarks: environmental selection in the consumption and miscellaneous sectors, social selection in the total universe or the consumption and financial sectors, and selection based on the company scores in the total universe or the financial and miscellaneous sectors.¹⁸ Not more than one case of significant underperformance (and thus no systematic effects) can be found when investors apply environmental screens in the capital and financial sector, social screens in the capital and miscellaneous sector, governance screens in the entire universe or consumption, financial and miscellaneous sectors, and screens

¹⁸ This suggests that IPS may gain a financial advantage by excluding high-rated stocks from the benchmark universe. However, we find that this approach does not create portfolios with statistically significant differences to the benchmarks.

Table 4

Performance of ESG-based selection for different cut-off rates.

	Asia-Pacific					United States					Europe				
	Total	Cap	Con	Fin	Misc	Total	Cap	Con	Fin	Misc	Total	Cap	Con	Fin	Misc
<i>High-to-BM</i>															
5% Environmental	–	–	–	–	–	+	–	+	+	–	c–	c–	–	–	b–
10% "	–	–	–	+	–	+	–	+	+	–	b–	–	b–	–	b–
15% "	–	–	–	+	–	+	–	+	+	–	b–	–	b–	–	b–
20% "	–	–	–	–	–	–	+	+	+	–	c–	–	–	–	b–
25% "	–	–	–	–	–	–	+	+	+	b–	–	–	b–	–	–
5% Social	–	–	+	–	–	–	–	+	+	+	–	–	–	–	–
10% "	–	–	+	–	–	–	–	+	+	–	c–	–	c–	b–	–
15% "	–	–	–	–	–	–	–	+	+	–	b–	–	b–	b–	–
20% "	–	–	–	–	–	–	–	+	+	–	b–	–	b–	b–	b–
25% "	–	–	–	–	–	–	–	+	+	–	b–	–	b–	b–	–
5% Governance	+	+	–	+	–	–	–	–	+	c–	c–	b–	c–	–	–
10% "	+	–	–	+	–	–	–	–	+	–	–	b–	–	–	–
15% "	+	+	–	+	+	–	–	–	+	c–	–	c–	–	–	–
20% "	+	–	–	+	+	–	–	–	+	–	–	–	–	–	–
25% "	+	–	–	+	+	+	–	–	+	–	–	–	–	–	–
5% Company	–	–	–	+	+	–	–	+	+	b–	c–	c–	–	–	–
10% "	–	–	–	+	+	+	–	+	+	–	b–	b–	–	c–	b–
15% "	–	–	–	+	+	+	–	+	+	–	b–	–	c–	c–	b–
20% "	–	–	–	–	+	–	–	–	+	c–	b–	–	–	b–	b–
25% "	–	–	–	–	+	–	–	–	+	–	b–	–	–	c–	b–
<i>Low-to-BM</i>															
5% Environmental	+	+	+	+	+	–	–	–	+	–	+	+	+	+	+
10% "	+	+	+	–	+	–	–	–	–	–	+	+	–	+	+
15% "	+	+	+	+	+	–	–	–	–	–	+	+	+	+	+
20% "	+	c+	+	+	+	–	–	–	–	+	+	c+	+	+	+
25% "	+	b+	+	+	+	+	–	–	–	+	+	+	+	+	+
5% Social	+	+	–	–	+	+	+	+	–	+	+	+	+	+	+
10% "	+	+	+	–	+	+	+	c+	–	+	+	+	+	+	+
15% "	+	+	+	–	+	+	+	+	–	+	+	+	+	+	–
20% "	+	+	+	–	+	+	+	+	–	+	+	+	+	+	+
25% "	–	+	–	–	+	+	+	+	+	+	+	+	+	+	+
5% Governance	+	–	–	–	–	–	–	–	–	–	–	+	+	–	+
10% "	–	–	+	–	–	–	–	–	–	–	+	+	+	–	–
15% "	–	–	–	–	–	–	–	–	–	–	+	+	+	+	–
20% "	–	+	+	–	–	–	–	–	–	–	+	+	+	+	–
25% "	–	+	+	–	–	–	–	–	–	–	+	+	+	+	–
5% Company	+	+	–	+	+	–	+	–	–	–	+	+	–	–	–
10% "	–	+	+	–	+	–	–	+	–	–	+	+	+	+	+
15% "	+	+	+	–	+	–	–	+	–	–	+	+	+	+	+
20% "	+	+	+	–	+	–	–	+	–	–	+	+	+	+	–
25% "	+	+	+	–	+	–	+	+	–	+	+	+	+	+	+
<i>Low-to-High</i>															
5% Environmental	+	+	+	+	+	–	+	–	–	+	+	+	+	+	+
10% "	+	+	+	–	+	–	+	–	–	+	+	+	+	+	b+
15% "	+	+	+	+	+	–	–	–	–	+	+	+	c+	+	+
20% "	+	c+	+	+	+	–	–	–	–	+	+	+	+	+	c+
25% "	+	c+	+	+	+	+	–	–	–	c+	+	+	+	+	+
5% Social	+	+	–	–	+	+	+	+	–	–	+	+	+	+	+
10% "	+	+	–	–	+	+	+	+	–	+	+	+	+	+	+
15% "	+	+	+	–	+	+	+	+	–	+	c+	+	c+	+	+
20% "	+	+	+	–	+	+	+	–	–	+	c+	+	c+	+	+
25% "	+	+	+	–	+	+	+	–	+	+	+	+	c+	c+	+
5% Governance	–	–	+	–	+	–	–	+	–	+	+	+	+	+	+
10% "	–	–	+	–	–	–	+	+	–	+	+	b+	+	+	+
15% "	–	–	+	–	–	–	+	+	–	+	+	c+	c+	+	+
20% "	–	+	+	–	–	–	+	–	–	+	+	+	+	+	+
25% "	–	+	+	–	–	–	+	–	–	–	+	+	+	+	+
5% Company	+	+	–	–	+	–	+	–	–	+	+	+	+	+	+
10% "	+	+	+	–	–	–	+	–	–	+	b+	b+	+	+	+
15% "	+	+	+	–	–	–	+	–	–	+	+	+	c+	+	+
20% "	+	c+	+	–	–	–	+	+	–	+	+	+	+	c+	+
25% "	+	+	+	–	–	–	+	+	–	+	+	+	+	c+	+

This table reports the signs (+ and –) and significances (a, b and c for levels of 1%, 5% and 10%, respectively) for the differences of portfolio Sharpe ratios, where the latter are evaluated on the basis of the LW test outlined in Section 3.2. High- and low-rated portfolios are compared to their benchmarks (BM). In addition, the differences between low-rated portfolios and their high-rated counterparts are analysed. The given percentages resemble the cut-off rates used in the ESG-screened portfolios. Grey shading indicates significant differences belonging to industry sectors with at least four significant cases of high-rated portfolios underperforming the benchmarks. Industry abbreviations are used as in Table 1.

based on the total company score in the consumption sector.¹⁹ In these segments, VDI tend to achieve performance similar to the benchmarks. As there is only one segment (20% environmental cut-off in the capital sector) where ESG-selected portfolios outperform the benchmark, neither RPS nor IPS can realise their investment goals, i.e. beat the market by active ESG-based stock selection. Finally, we can identify 17 cases where low-rated portfolios significantly outperform their high-rated counterparts. However, the outperformance is not strong enough to simultaneously outperform the benchmarks.

4.2. Robustness checks

As our main results are subject to several issues of arbitrary choice, this section describes a variety of supplementary calculations that verify the robustness of our results. The most important of these robustness checks considers the possibility that our results are just a sample specific phenomenon. To check the validity of our results over time, we perform the following calculations. Each month in our sample, we estimate the Sharpe ratios of our portfolios by using the excess return data of that month and all previous months. Because Fung and Hsieh (1997) and Liang (2000) emphasise that the minimum sample size for obtaining meaningful performance measures is 36 months, our computations start in August 2007. Fig. 2 exemplarily plots the resulting Sharpe ratios for the 20% high and low cut-off portfolios belonging to the European 'financially harmful' segments where VDI appear to pay a price (reduced risk-adjusted performance) for being socially responsible.²⁰

From the graphs, we can deduce four important implications. First, high-rated portfolios that have been identified to significantly underperform the passive non-ESG strategies show Sharpe ratios that are persistently below the one of the benchmark.²¹ Second, the differences of underperforming portfolios to the benchmark become larger beginning with the year 2009. Supported by subsequent application of the LW test, before 2009 some of these underperformers show performance that does not significantly differ from the benchmark (e.g. the portfolio resulting from a screen in the entire universe that is based on the company score). Third, low-rated portfolios show performance very similar to the benchmark. For two exceptions (social and company screens in the financial sector), we can observe Sharpe ratios somewhat higher above the benchmark. However, the differences are not significant according to the LW test. Fourth, the consistently highest underperformance can be observed for the environmental screen in the miscellaneous sector.

After analysing the robustness of our results over time, we consider seven additional robustness checks that deal with the estimation of alternative alphas, the specification of our bootstrap-based testing procedure, the portfolio construction, the choice of risk-free rate, the preparation of the dataset and the consideration of transaction costs:²²

¹⁹ Note that the signs of the Sharpe ratio differences in the governance screens for the United States are noticeable different from the European ones. This partially resembles the effects of the different corporate governance systems in the United States and Europe (see Shleifer & Vishny, 1997).

²⁰ The other cut-off rates that have proven to yield significant performance differences show similar results. Furthermore, insignificant segments do not show remarkable differences over time.

²¹ Similar evidence on the persistence of relative performance can be obtained when shifting a rolling window of constant 36 months over the sample period. Thus, the differences in performance are not driven by initial strong underperformance followed by similar development in the subsequent months.

²² As these calculations do not provide any additional valuable insights besides showing results similar to Table 4, we concentrate on verbally summarising their main implications. Detailed results can be made available upon request.

First, for reasons of completeness, we follow Bauer et al. (2005) and also estimate the alphas of our SRI portfolios using multi-factor models that contain classic small-minus-big, high-minus-low and momentum factors. For each geographical region, the factor portfolios are constructed following Artmann et al. (2012).²³ Interestingly, the alphas identify the same 'financially harmful' segments as our Sharpe ratio analysis. Additionally, differences between the alphas of low-rated and high-rated portfolios carry the same signs. In the insignificant segments, results differ only in a negligible number of cases.

Second, we use alternative block sizes between 1 and 15 (the largest reasonable block size implemented in the programme codes of LW) in our bootstrap test and apply the calibration procedure of LW. However, as indicated by Auer and Schuhmacher (2013), this does not crucially influence the test outcomes in our application.

Third, even though several studies show that no alternative classic approach is consistently better than the naive 1/n weighting scheme (see DeMiguel, Garlappi, & Uppal, 2009; Tu & Zhou, 2011), we follow Yao (2012) by also examining value-weighted returns because this prevents the results from being driven by smaller, less liquid firms. However, our conclusions turn out to be similar to those obtained for the equally-weighted portfolios.

Fourth, as in Frazzini and Pedersen (2014), alternative choices of risk-free rates are used for the calculation of excess returns. A constant risk-free rate equal to the average interest rate for the period under consideration and the interest rate at the beginning of the investigation period is employed. Furthermore, the interbank offered rates are exchanged by Datastream benchmark bond yields broken down to a monthly equivalent. All approaches yield almost identical results.

Fifth, to avoid outlier-biases, some studies suggest the exclusion of penny stocks (see Ince & Porter, 2006), the 5% of firms with the lowest market capitalisation in each region (see Ang, Hodrick, Xing, & Zhang, 2009; Walkshäusl, 2014) or the 0.25% smallest and largest return observations (see Artmann et al., 2012). As our sample does not contain penny stocks, we follow the two latter suggestions and find that our conclusions still hold.

Sixth, we employ an alternative portfolio construction methodology as used in the standard empirical asset pricing literature (see Goyal, 2012). Specifically, we build quintile (decile) portfolios and compare the highest quintiles (deciles) to the corresponding lowest (deciles) and compare both to our benchmarks. Furthermore, we also create a portfolio that goes long in the portfolio with the highest ESG ratings and short in the portfolios with the lowest ESG ratings (for details, see Kempf & Osthoff, 2007), test the significance of its returns and compare the portfolio to our benchmarks. In all cases, our overall conclusions remain valid.

Finally, we consider transaction costs (the costs of forming the portfolios in August 2004, adjusting the portfolios within the investment period, and closing the portfolios in December 2012) by following Derwall et al. (2005) and Lucey and Zhao (2008) by assuming round-trip transaction costs between 50 and 200 basis points. As in Kempf and Osthoff (2007), we find that this does not alter our main conclusions regarding the benefits of ESG-based portfolio selection.²⁴ An explanation for this can be found by taking a closer look at the changes of the ESG ratings over time. If changes do not occur very often, portfolio decompositions do not have to be adjusted regularly and, thus, transaction costs do not play a vital role. In the Asia-Pacific region, environmental, social,

²³ In an additional robustness check, we also use the geographic factors of the Frazzini dataset available under www.aqr.com/library/data-sets/betting-against-beta-economy-factors-monthly.

²⁴ Similar results can be obtained for the computation of break-even transaction costs as in Della Corte, Sarno, and Tsiakas (2009).

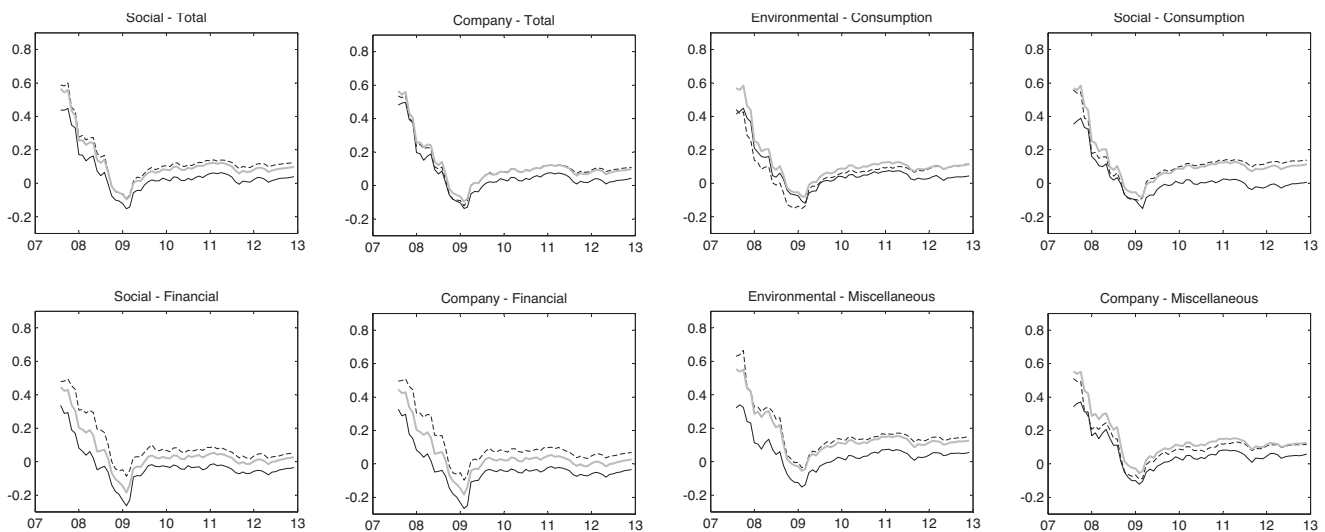


Fig. 2. Sharpe ratio performance in 'financially harmful' segments over time. For the European high and low portfolios with a 20% cut-off rate belonging to the identified 'financially harmful' segments, this figure shows the evolution of the Sharpe ratios over time. Each month (starting in August 2007), the Sharpe ratio is calculated by using the excess return data of that month and all previous months. The solid grey, solid black and dashed black lines resemble the Sharpe ratios of the benchmark, the high-rated and the low-rated portfolios, respectively.

governance and company scores on average change in 23.00%, 18.05%, 15.05% and 24.00% of the time periods, respectively.²⁵ In the United States and Europe, we get 20.46%, 16.73%, 13.99%, 22.64% and 23.35%, 20.25%, 17.60%, 24.79%, respectively. Thus, unlike portfolio construction based on market capitalisation, for example, ESG-based strategies have lower adjustment frequencies and are thus exposed to lower (largely negligible) transaction costs (see also Baker, Bradley, & Taliaferro, 2014). Furthermore, the low levels of these average changes in combination with the rather low standard deviation of the ESG scores (as it can also be observed in the ESG data of other data vendors) provide an explanation for the fact that our results are also robust to changes of the portfolio rebalancing frequency from monthly to annually or even biennially.

5. Conclusion

Given the growing social awareness of investors and the increasingly positive regulatory environment (see Renneboog et al., 2008a), investment strategies based on measures of CSR have become a major topic of interest for both researchers and practitioners. Initial empirical evidence from the mutual fund literature showed that there are no significant performance differences between SRI funds and conventional funds. Follow-up results from direct portfolio construction that can deal with some problems of investment fund analysis (e.g. the fact that funds' returns also reflect management fees) provided mixed results that are not strictly comparable because of the focus on specific geographical regions or dimensions of CSR, different measures of CSR, high aggregation or other drawbacks. By using a new international dataset of Sustainability ESG scores, breaking the aggregation on the ESG and the country level by using ESG subscores and industry subsets and applying a state-of-the-art statistical performance measurement approach based on total risk, we seek to mitigate those drawbacks and provide new and robust evidence on whether socially

responsible or irresponsible stock selection can outperform passive non-ESG investments in international equity markets.

We find that the geographic and industry focus of an ESG-based investment strategy and the employed ESG criteria strongly determine its outcome. In the Asia-Pacific region and the United States, selecting high (low) ESG stocks does not appear to consistently increase or decrease investment performance relative to the benchmarks and to low (high) ESG stocks. In Europe, we also find no evidence for the superiority of ESG-based strategies. On the contrary, in certain industries and depending on the ESG criterion, investors pay a price for being socially responsible in their stock selection, i.e. they end up with significantly lower risk-adjusted performance than the passive benchmarks. Thus, in summary, our findings rehabilitate previous research on the fund level and fully support its main results.

As private investors typically evaluate the performance of investment funds by comparing them to broad stock market indices (like our benchmark strategies) our results have important implications for the construction and marketing of ethical funds. Fund managers can satisfy the ethical needs of their investors by ESG-based stock selection but at best provide a performance similar to the broad market. While in the Asia-Pacific region and the United States this can be achieved regardless of the ESG-criterion or industry focus, in Europe some combinations of ESG-criteria and industries should be avoided so as not to cause financial disadvantages. Thus, fund managers have a variety of settings at their disposal in which they can satisfy the needs of value-driven investors by offering 'socially filtered' market performance. The needs of profit-seeking investors cannot be met by ESG-based stock selection – either by choosing socially desirable or by socially less desirable stocks.

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²⁵ These percentages are obtained as follows: For each stock in a geographic region, we count the number of changes in the ratings and divide them by the number of total observation periods. The cross-sectional average of the resulting ratios is then multiplied by 100 to yield the stated frequencies.

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